

Dr. JUTAN DAS

Ph.D. (Bioinformatics), ICAR-IARI & ICAR-IASRI, New Delhi
Assistant Professor | Bioinformaticist | AI/ML & Computational Biology Researcher
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RESEARCH INTERESTS

· Bioinformatics · Agricultural Biotechnology · Artificial Intelligence & Machine Learning · Deep Learning · Computational Genomics · CRISPR-Cas9 · Non-Coding RNAs · NGS · Transcriptomics · Bioinformatics Database & Web Tool Development · Agricultural Statistics.

ACADEMIC ACHIEVEMENTS

- All India Rank (AIR) 4 — **IARI Ph.D. Entrance Examination** (Bioinformatics), 2019.
- UPS Rank 1st and Category Rank 3rd - **ICAR-AIEEA (PG) Examination**, 2017.
- **ICAR-IARI Ph.D. Fellowship** Awardee (2019–2025) — Awarded for Ph.D. (Bioinformatics) at ICAR-IARI, New Delhi.
- **ICAR-IARI M.Sc. Fellowship** Awardee (2017–2019) — Awarded for M.Sc. (Bioinformatics) at ICAR-IARI, New Delhi.
- 🏆 **Best Research Paper Award** — Presented at the 3rd International Conference on 'Innovative Approaches in Agriculture, Horticulture & Allied Sciences' (IAAHAS 2023), SGT University, Gurugram (ICAR Accredited). March 29–31, 2023.
- 🏆 **Best Poster Presentation Award** — Presented at the International Conference on 'Current Innovations and Technological Advances in Agriculture and Allied Sciences' (CITAAS-2024), Guru Kashi University, Talwandi Sabo (ICAR Accredited). August 29–31, 2024.
- Software **Copyright** Holder- *CCncRNadb: Cyprinus Carpio ncRNAs Database*. Govt. of India Reg. No. 26561/2024-CO/SW.

PERSONAL INFORMATION

Date of Birth : 2nd August, 1993
Gender : Male
Nationality : Indian
Marital Status : Unmarried
Languages : English, Bengali, Hindi

Permanent Address: Khas Chowmuhani, Melaghar, Sepahijala, Tripura – 799115
Current Address: Block-5, Sector-27, Atal Nagar-Nava Raipur, Chhattisgarh – 492018
Hobbies: Diary Writing, Dancing, Gardening, Cricket

ACADEMIC QUALIFICATIONS

Degree	Board / University	Year	Marks (%)	Division
Ph.D. (Bioinformatics)	ICAR-Indian Agricultural Research Institute, New Delhi	2025	80.10%	First
M.Sc. (Bioinformatics)	ICAR-Indian Agricultural Research Institute, New Delhi	2019	79.50%	First
B.Sc. (Agriculture)	College of Agriculture, Tripura (Lembucherra)	2017	78.50%	First

PROFESSIONAL EXPERIENCE

Assistant Professor (Bioinformatics) | February 2026 – Present

Department of Bioinformatics, Kalinga University, Raipur, Chhattisgarh

Teaching Responsibilities

- Teaching core Bioinformatics courses at UG, PG, and Ph.D. levels, including: Programming for Biologists (Python, R, SQL), Genomics & Proteomics, Molecular Modeling & Drug Design, Machine Learning & Deep Learning in Biology, Biostatistics, Sequence Analysis, and Structural Bioinformatics.
- Designing course curricula, lecture notes, practical manuals, and continuous assessment frameworks.
- Supervising student research projects, dissertations, and Ph.D. thesis work.
- Mentored students in manuscript preparation; co-authored student research and review articles.

Administrative & Academic Service Responsibilities

- Member, Board of Studies (BoS) — Department of Bioinformatics, Kalinga University, Raipur.
- Organising Committee Member — Two-Week Online Faculty Development Programme (FDP) on 'Integrating Science and Technology for Sustainable Development and Global Transformation', organised by the Department of Physics & the Department of Bioinformatics, Kalinga University, in association with CGCOST Raipur & Siddhachalam Lab, Raipur. June 15–26, 2026.
- Serving as Departmental Training & Placement (T&P) Coordinator.
- Contributing to academic administration as Magazine Editor and Cultural Event Coordinator.

RESEARCH EXPERIENCE

A. M.Sc. Research

Computational Intelligence in the Estimation of CRISPR-Cas9 Cleavage Sites — Developed advanced computational models and algorithms to predict CRISPR-Cas9 off-target sites with enhanced precision and efficacy for plant genome editing applications.

B. Ph.D. Research

Development of Artificial Intelligence-Based Fish-Specific Long Non-Coding RNA Biomarker Discovery Tool and Web Genomic Resource — Built AI/ML/DL-based models to accurately predict, identify, and characterize lncRNAs in fish species; developed associated databases and web servers for the scientific community.

C. Tools, Databases & Web Servers Developed

1. CCncRNAdb — Database for *Cyprinus carpio* lncRNAs and circRNAs (Copyright Reg. No. 26561/2024-CO/SW). <http://backlin.cabgrid.res.in/ccncrnadb/>
2. FishLncPred — ML/DL-based web server for lncRNA prediction in fish species. <http://46.202.167.198:5000/>
3. LncSubLoc — Web server for lncRNA sub-cellular localization prediction. <http://login1.cabgrid.res.in:5200/>
4. i-CRISPR — ML-based classifier for CRISPR-Cas9 on/off-target prediction in plant genomes. <http://login1.cabgrid.res.in:5092/>
5. CyExpDB — Comprehensive gene expression atlas for the Cyprinidae family. <http://backlin.cabgrid.res.in/cyexpdb/>
6. Developed ANN, SVM, RF, CNN, and FNN models for CRISPR-Cas9 off-target site prediction in plant systems.
7. Developed an AI/ML model to predict long non-coding RNAs (lncRNAs) in fish using a curated dataset with advanced feature extraction and selection techniques.

RESEARCH PUBLICATIONS

A. Published Research Articles

1. **Das, J.**, Kumar, S., Mishra, D. C., Chaturvedi, K. K., Paul, R. K., & Kairi, A. (2023). Machine learning in the estimation of CRISPR-Cas9 cleavage sites for plant system. *Frontiers in Genetics*, 13, 1085332.
2. **Das, J.**, Kumar, B., Saha, B., Jaiswal, S., Iquebal, M. A., Angadi, U. B., & Kumar, D. (2023). Genome-wide identification and characterization of tissue specific long non-coding RNAs and circular RNAs in common carp (*Cyprinus carpio* L.). *Frontiers in Genetics*, 14, 1239434.
3. Saini, P., Chaudhary, K., **Das, J.**, Kumari, N., Jaiswal, S., Rasal, K. D., & Kumar, D. (2025). CyExpDB: a web-based multi-species tissue-specific gene expression platform for functional genomics in Cyprinidae fish. Database, 2025, baaf087.

B. Accepted Articles

1. Navratna, D. K., & **Das, J.** (2026). Graph neural networks for antimicrobial drug discovery: A systematic review of trends, gaps, and explainability challenges. *Univista: International Journal of Research and Studies*, 1(2). (Accepted)
2. Kanwar, S., & **Das, J.** (2026). Deep learning-based CRISPR-Cas9 off-target prediction: A critical review of existing models and their limitations for plant genome editing applications. *Univista: International Journal of Research and Studies*, 1(2). (Accepted)

C. Manuscripts Under Review / Submitted

1. **Das, J.**, Saha, B., Saini, P., Jaiswal, S., Iquebal, M. A., Angadi, U. B., & Kumar, D. (2026) FishLncPred: An Advanced Machine Learning Tool for Predicting Fish lncRNAs. *Aquaculture*. (Submitted).
2. Chandini, B. C., **Das, J.**, and Kumar, S. (2026). Deep learning-based approach for prediction of CRISPR/Cas9 off-target sites in plant systems. *Current Bioinformatics*. (Under Review).

3. Kumar, B., Sharma, N. K., **Das, J.**, Iquebal, M. A., Jaiswal, S., Angadi, U. B., Kumar, D. (2026). lncRNA sub-cellular localisation prediction using multi-source features. *Current Bioinformatics*. (Under Review).
4. Anand, A., **Das, J.**, Kumar, S., Mishra, D. C., Srivastava, S., Varghese, C., Chandini, B. C., Ravi., Sharma, N. K. (2026). i-CRISPR: Machine learning based classifier for CRISPR-Cas9 on/off-target. *Current Genetics*. (Submitted).
5. Samal, I., Mahanta, D. K., Choudhary, J. S., Bhoi, T. K., Komal, J., Murmu, S., Bharti, S. K., Tanvir, M., & **Das, J.** (2026). Insects as Sentinels of Environmental Change: Morpho-Chemical Biomarkers and One Health Implications. *Agricultural Environment and Sustainability*. (Under Review).

D. Popular Articles

1. Das, J. (2024). Prediction of lncRNAs using machine learning-based tools. *Food and Scientific Reports*, 5(4): 6–10.

E. Copyright

1. Obtained copyright for “CCncRNadb: *Cyprinus Carpio* ncRNAs Database” as computer software. Registration No. 26561/2024-CO/SW.

CONFERENCE PRESENTATIONS

1. **Das, J.** & Kumar, S. (2022). *Machine Learning in the Estimation of CRISPR-Cas9 Cleavage Sites*. Paper presented at the 73rd Annual Conference of ISAS on Statistics and Machine Learning for Big Data Analytics, SKUAST-K, Srinagar, J&K, India. November 14–16, 2022.
2. **Das, J.**, Jaiswal, S. & Iquebal, M. A. (2023). *Identification and Characterization of Tissue-wise Non-coding RNAs in Common Carp (*Cyprinus carpio* L.)*. Paper presented at the 44th Annual International Conference on Recent Trends in Statistics & Data Analytics (RTSD-2023), Mar Ivanios College, Thiruvananthapuram, India. March 31 – April 1, 2023.

TECHNICAL SKILLS

Programming & Computing

- Python, R, Perl, BioPerl, SQL, Shell Scripting
- Linux, HPC
- Flask, Jupyter Notebook, PyCharm

Machine Learning & Deep Learning

- Algorithms: RF, SVM, ANN, KNN, DT, XGBoost
- Deep Learning: CNN, FNN - TensorFlow, Keras
- Libraries: Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Statsmodels

Bioinformatics

- CRISPR-Cas9 off-target analysis
- RNA World: lncRNA and circRNA
- Genomics and Transcriptomics

Database & Web Development

- Bioinformatics database design and deployment
- Web server development for biological data analysis
- Database and web server visualization

General

- Student mentoring and curriculum development
- Scientific writing and academic publishing
- Microsoft Word, Excel, and PowerPoint

NGS & Transcriptomics

- RNA-Seq data analysis
- Transcriptome data mining
- Gene Expression Analysis
- Functional annotation

TRAININGS & WORKSHOPS

1. 3-Day Training on '*Next Generation Sequencing and Computational Biology for Agriculture*' — ICAR-IARI, New Delhi & University of Colorado Boulder, USA (CAAST-NAHEP, World Bank Funded). March 4–6, 2020.
2. International Symposium on '*Advances in Plant Biotechnology and Genome Editing*' & 42nd Meeting of Plant Tissue Culture Association (India) — APBGE-2021, ICAR-IIAB, Ranchi, India. April 8–10, 2021.
3. Certification: Python for Machine Learning & Deep Learning — Great Learning Academy (2020).

STRENGTHS

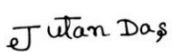
- Advanced expertise in AI/ML/DL model development for biological and agricultural applications.
- Strong computational biology research background with proven publication record in high-impact journals.
- Ability to design and deploy end-to-end bioinformatics tools, databases, and web servers.
- Effective scientific communicator with experience in teaching, mentoring, and academic administration.

DECLARATION

I hereby declare that all information stated above is true, complete, and accurate to the best of my knowledge.

Date: 9th July, 2026

Place: Raipur


Dr. JUTAN DAS

PROFESSIONAL REFERENCES

1. Dr. Sarika Jaiswal

Principal Scientist & Professor, Division of Agricultural Bioinformatics, ICAR-IASRI, New Delhi

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2. Dr. Mir Asif Iquebal

Principal Scientist, Division of Agricultural Bioinformatics, ICAR-IASRI, New Delhi

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3. Dr. Dinesh Kumar

Principal Scientist, Division of Agricultural Bioinformatics, ICAR-IASRI, New Delhi

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